

Fighting the **FLU** The bad news: H1N1 flu is highly contagious and has spread to 168 countries



INDIA

Casualty:
17 deaths, 1,000-plus infected.

Economic cost:

The GDP may take a hit of 1-1.5%

Battle plan:
Entry screening at air and sea ports; contact tracing of people who may have been infected, more testing laboratories, stocking up on Tamiflu.



US

Casualty:
436 deaths and over 50,000 positive.

Economic cost:

A drop of 0.5-1.5% expected in GDP

Battle plan:
Vaccination and communications programmes. Decisions about school closures left to local authorities. \$7.7 b funding likely.



UK

Casualty:
40 deaths, around 12,470 positive.

Economic cost:

GDP may be hit by 3%

and where. The H1N1 flu is astonishingly contagious, but can linger beneath the radar of health monitors with ease. "With SARS and Avian flu, patients showed symptoms as they got infected. But with H1N1, people can spread infection days before they feel sick enough to go to a doctor," says Guleria. It's also unpredictable, because adults older than 64 years do not appear to be at risk.

In the US, the numbers shot up rapidly and when its Centre for Disease Control decided to stop listing the number infected on July 24, those who had tested positive for the flu totalled 43,771 of which 302 deaths had been reported. The death toll since then has risen to 436. WHO estimates that worldwide the number infected was 1,77,457 on August 12 and the death toll was 1,462.

While H1N1 virus worldwide is threatening to reach pandemic proportions that WHO had warned about, in terms of virulence and mortality it is far milder than feared. Worldwide the mortality rate is still low at around 0.8 per cent (or eight out of 1,000 people infected with H1N1 will die). Despite the toll shooting up last week, mortality rates in India are at par with those in other countries. The rate, it pays to remember, is lower than that of deaths due to tuberculosis, diarrhoea and pneumonia.

Yet, in India the panic has been so widespread that it has caught Central and state Governments flatfooted.

With just 19 testing laboratories, a stock of 10 million antivirals, no vaccination and little support from private players who control 70 per cent of the health sector, the odds seem to be heavily stacked against the Government. Young Reeda Shaikh's death in Pune may have galvanised the Government into action, but is India ready to handle a flu that has already spread to more than 22 cities and towns across the country?

Going by the non-resident Gujarati Pravin Patel—who died of H1N1 last week, within 10 days of landing in Ahmedabad from Atlanta—probably not. Patel's family still blames the medical team at the airport who had failed to catch his condition at their customary screening. "We have now declared it as a national emergency because we really don't know ultimately what the size of the seriousness of this outbreak is going to be and we don't want to be unprepared or found wanting," says Union Health Minister, Ghulam Nabi Azad (see interview).

In a way the influenza epidemic was waiting to happen. Going by past records, one influenza pandemic takes place every four decades, writes researcher on medical history, Eugenia Tognotti in the *Journal of Infection in Developing Countries* in May 2009. India faced two major pandemics in the last century: the Spanish flu of 1918-20 and the Asian flu pandemic in 1957-60. The former had killed between 50 million and 100 million people globally (4,000 per million in India). The latter, though milder, had

What's new about **H1N1**?



It's a new virus that combines genes from birds, pigs and humans—something scientists have never seen before. Though it's behaving like any influenza virus nobody has any immunity to it. A 'Type A', it can replicate, infect and spread easily from human to human.

Is there a **VACCINE**?



Because the virus is new, there is no vaccine ready to protect against pandemic flu. A vaccine against swine flu virus H1N1 needs at

least 6-12 months to be brewed. Seasonal flu vaccine or past flu immunisation will not provide protection.

How effective are **MASKS**?



Apart from the N95 respirator (considered the best with 95 per cent protection), most other masks available in the market are not effective.